Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-9550
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies used to build and configure the Script-Controlled ACL solution. The stack is centered on the ServiceNow platform, server-side scripting, ACL configuration, and the platform data layer.

Fill out the table below to detail the technologies chosen for each architecture component and provide a brief justification for their use.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	**ServiceNow UI** Forms & Lists	Provides the user interface for viewing and working with institution/student records. The project verifies ACL behavior through ServiceNow forms and lists.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	**ServiceNow Server-Side JavaScript** ACL Script	The project uses an ACL script to evaluate roles and control access. The script allows administrators, permits the required custom-role access, and denies other users.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	**ServiceNow Table / Data Layer** u_institution_details	Stores the project records and fields such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description. ACLs protect access to these records.

4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	**ServiceNow Cloud Platform**	The solution is configured and executed inside the ServiceNow instance. This avoids a separate application server for the ACL implementation and keeps security enforcement within the platform.
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	**ServiceNow Update Sets**	Supports moving configuration changes such as tables, roles and ACL definitions between ServiceNow instances. It is suitable for managing configuration changes for this platform-based project.
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	**ServiceNow ACL Engine & Security Admin**	No external third-party API is required for the core solution. The ACL engine, custom roles, Security Admin elevation, and impersonation/testing features provide the required access-control workflow.

Project reference: The technology choices above are aligned with the supplied project document, which defines a ServiceNow custom table, custom roles, and script-controlled READ, CREATE, WRITE and DELETE ACLs.